HOL3: Experiment Orchestrator - Ship/No-Ship Decision Memo

Date: 2026-10-04 | Model: gpt-4o-mini | Cases per variant: 25

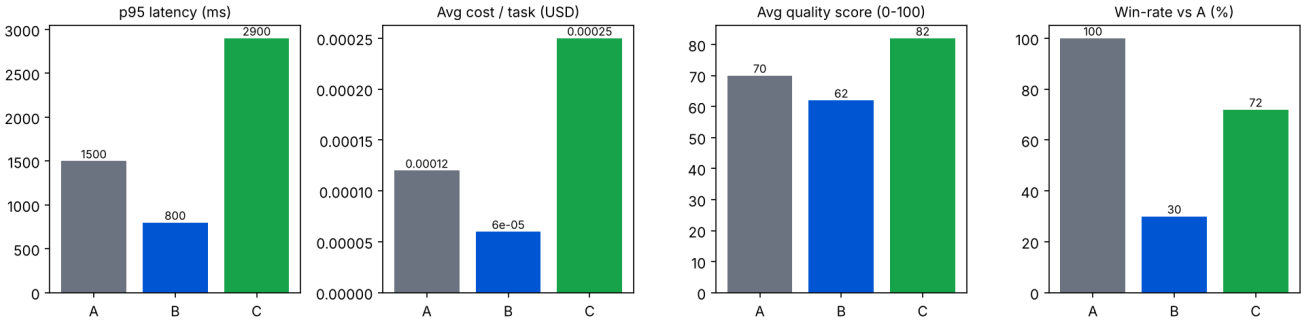
SAMPLE LAYOUT - SYNTHETIC MOCK DATA, NOT A REAL EXPERIMENT. Do not submit.

Recommendation

NO-SHIP: keep Variant A (Baseline)

Metrics

Metric	A Baseline	B Fast	C High-Quality
p50 latency (ms)	900	500	1600
p95 latency (ms)	1500	800	2900
Avg tokens/sec	80.0	85.0	90.0
Avg cost/task (USD)	0.000120	0.000060	0.000250
Avg quality (0-100)	70.0	62.0	82.0
Win-rate vs A (%)	100	30	72
Error rate (%)	0.0	0.0	0.0



Justification and trade-offs

Neither B nor C met every ship criterion against baseline A: B failed [quality case]; C failed [p95 <= 1.5x A, cost <= 1.5x A].

Consistency (p95/p50): A 1.67x, B 1.60x, C 1.81x. Fastest p95: B; cheapest: B; highest quality: C.

Speed vs quality: B cuts p95 by 47% but its quality score is -8.0 points vs A; C scores +12.0 points but its p95 is +93% vs A.

Cost vs quality: per-task cost is -50% for B and +108% for C relative to A, so any quality gain from C is bought with higher spend and longer tails.

Quality here is a heuristic (length, structure, keywords), not human or LLM-judge evaluation, so win-rate favours longer answers and should be treated as directional.

Rollback trigger

Within the first 24 hours, roll back to Variant A if p95 latency exceeds 3000 ms (2x baseline), error rate exceeds 2%, or cost per task exceeds $0.000180 (1.5x the measured value).

Next step

Replace the heuristic with an LLM-as-judge or human review on a sample, add repeat runs to estimate variance, and test a hybrid that routes simple prompts to the cheaper variant.